

Experiment.4

Aim: Design and Implementation of an Obstacle Avoiding Car Using Arduino

Introduction:

An **Obstacle Avoiding Car** is an autonomous robotic vehicle designed to detect and avoid obstacles in its path without human intervention. The system uses an **Arduino Uno** as the central controller, an **ultrasonic sensor** to measure distance, and a **motor driver** to control the movement of DC motors attached to the car's wheels. By continuously monitoring the environment, the car adjusts its motion in real-time, ensuring smooth navigation and collision avoidance.

This project demonstrates the integration of sensor input, motor control, and decision-making using Arduino. It provides a practical foundation for understanding autonomous systems and real-time embedded programming.

Arduino Uno Overview

The **Arduino Uno** is a popular microcontroller board known for its simplicity, low cost, and ease of use. It features:

- **Microcontroller:** Atmega328P
- **Operating Voltage:** 5V
- **Digital I/O Pins:** 14 (6 PWM outputs)
- **Analog Input Pins:** 6
- **Flash Memory:** 32 KB
- **Clock Speed:** 16 MHz
- **USB Connection:** For programming and communication
- **Power Supply:** Can be powered via USB or an external power supply

The Arduino Uno can be programmed using the **Arduino IDE**, which allows users to write and upload code to control electronic components connected to the board.

Components Used in This Experiment

1. Arduino Uno

- Acts as the central processing unit.
 - Processes sensor readings and controls the motors.
 - Makes decisions based on real-time sensor input.
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2. DC Motors (300 RPM)

- Power the wheels of the car.
 - The 300 RPM speed allows for controlled movement.
 - Motors enable forward, backward, and turning motions based on Arduino signals.
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3. Ultrasonic Sensor (HC-SR04)

- Detects obstacles by emitting sound waves and measuring the time it takes for them to return after hitting an obstacle.
- Calculates the distance to the object using the formula:

$$\text{Distance} = \frac{\text{Time} \times 0.0344}{2}$$

- If the distance is below a threshold (e.g., 20 cm), the Arduino makes a decision to avoid the obstacle.
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4. L293D Motor Driver

- Controls the direction and speed of the motors.
 - Receives control signals from the Arduino and adjusts power supply to motors.
 - Enables forward, backward, and turning movements.
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5. Wheels

- Attached to the motors, allowing the car to move in various directions.
 - The size and type of wheels affect traction and speed.
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How the Obstacle Avoiding Car Works

1. Sensor Input

- The ultrasonic sensor sends out sound waves using the **trigger pin** and listens for the return signal on the **echo pin**.
 - The Arduino measures the time taken for the pulse to return and calculates the distance.
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2. Decision Making

- If the distance is greater than the threshold (safe distance), the car continues to move forward.
 - If the distance is less than the threshold (obstacle detected), the Arduino takes action to avoid the obstacle (e.g., stopping or turning).
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3. Motor Control

- The L293D motor driver receives signals from the Arduino to adjust the motors' speed and direction:
 - **Move Forward:** Both motors rotate in the same direction.
 - **Turn Left/Right:** One motor rotates while the other remains stationary.
 - **Stop:** Both motors stop.
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4. Continuous Loop

- The car continuously monitors its environment and adjusts its movement based on real-time feedback.
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Simple Operational Flow

1. **Start:**

- System initializes; Arduino and sensors are powered on.

2. **Measure Distance:**

- Ultrasonic sensor sends out a pulse and measures the time for the echo to return.
- Arduino calculates the distance.

3. Obstacle Detection:

- If the distance < 20 cm $\rightarrow$ Stop and turn.
- If the distance > 20 cm $\rightarrow$ Continue moving forward.

4. Repeat:

- Continuously monitor and react to environmental changes.

Code Explanation

Arduino Code for Basic Obstacle Avoidance

cpp

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```
#define TRIG_PIN 9
#define ECHO_PIN 10
#define MOTOR1_PIN1 5
#define MOTOR1_PIN2 6
#define MOTOR2_PIN1 7
#define MOTOR2_PIN2 8

void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(MOTOR1_PIN1, OUTPUT);
  pinMode(MOTOR1_PIN2, OUTPUT);
  pinMode(MOTOR2_PIN1, OUTPUT);
  pinMode(MOTOR2_PIN2, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  long duration, distance;

  // Send pulse from trigger pin
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
```

```
// Measure time for the pulse to return
duration = pulseIn(ECHO_PIN, HIGH);

// Calculate distance
distance = duration * 0.0344 / 2;

// Print distance to Serial Monitor
Serial.print("Distance: ");
Serial.println(distance);

if (distance < 20) { // Obstacle detected
    stopMotors();    // Stop the car
    delay(500);
    turnLeft();      // Turn the car left
    delay(500);
} else {
    moveForward();   // Move forward
}
}

void moveForward() {
    digitalWrite(MOTOR1_PIN1, HIGH);
    digitalWrite(MOTOR1_PIN2, LOW);
    digitalWrite(MOTOR2_PIN1, HIGH);
    digitalWrite(MOTOR2_PIN2, LOW);
}

void stopMotors() {
    digitalWrite(MOTOR1_PIN1, LOW);
    digitalWrite(MOTOR1_PIN2, LOW);
    digitalWrite(MOTOR2_PIN1, LOW);
    digitalWrite(MOTOR2_PIN2, LOW);
}

void turnLeft() {
    digitalWrite(MOTOR1_PIN1, LOW);
    digitalWrite(MOTOR1_PIN2, LOW);
    digitalWrite(MOTOR2_PIN1, HIGH);
    digitalWrite(MOTOR2_PIN2, LOW);
}
```

Code Breakdown

1. **Pin Setup:**

- Pins for motors and ultrasonic sensor are configured as input or output.

2. **Measuring Distance:**

- Pulse is sent and the return time is measured.
- Distance is calculated based on time taken.

3. **Decision Making:**

- If the distance < 20 cm $\rightarrow$ Stop and turn.
- If the distance > 20 cm $\rightarrow$ Continue moving forward.

4. **Motor Control:**

- `moveForward()` $\rightarrow$ Moves the car forward.
- `turnLeft()` $\rightarrow$ Turns the car left.
- `stopMotors()` $\rightarrow$ Stops the car.

Conclusion:

The Obstacle Avoiding Car is a simple yet effective autonomous system. By using an Arduino Uno, ultrasonic sensor, and motor driver, the car can detect obstacles and navigate around them by controlling the motors. The system continuously monitors its surroundings and makes real-time decisions to avoid collisions. This project is an excellent introduction to robotics and embedded systems, offering a solid foundation for building more complex autonomous vehicles in the future.